

2025 JC1 Promo Exam Paper 1 (Theory) – Suggested Solutions

1(a)	The first element in the array is considered a length 1 sorted array, 25. The remaining elements are considered unsorted. Compare the first unsorted element 54 with the elements in the sorted array 25. As 54 is bigger than 25, there is no shift/swap. The sorted array now grows to 25, 54 Next, consider element 33 to insert into the sorted array. 33 is not bigger than 54, so we shift 54 to the right. 33 is bigger than 25, so we do not shift 25. 33 is inserted in 25's right. The sorted array is now 25, 33, 54. Repeat inserting unsorted elements into the sorted array: 65 being the largest, results in no shift. The sorted array is now 25,33,54,65. Similarly, 73 is the largest, the sorted array is now 25,33,54,65,73 Similarly, 87 is the largest, the sorted array is now 25,33,54,65,73,87 With the next element 80, it will be inserted after 73 to become 25,33,54,65,73,80,87 With the next element 82, it will be inserted after 80 to become 25,33,54,65,73,80,82,87 which is the final sorted result.	1m - initial sorted array 1m - insertion (no shift) 1m - insertion (shift) 1m - correct middle to end state
(b)	Insertion sort has a time efficiency of $O(n^2)$ Merge Sort has a time efficiency of $O(n \lg n)$ Therefore, Merge Sort is more suitable for sorting large unsorted arrays.	1m – State $O(n^2)$ and $O(n \lg n)$ 1m – Conclude $n \lg n$ is better

2(a)	2725 has the hash of $2+7+2+5 \bmod 10 = 6$ so it will be inserted at index 6. However index 6 is occupied, so linear probing algorithm will check if index 7 is free, which it is. 2725 will be placed at index 7.	1m – hash 1m – use linear probing algorithm
(b)	The keys and the respective hashes are: 6201:hash 9 2352:hash 2 3857:hash 3 8131:hash 3 5731:hash 6 4249:hash 9 Since 2352, 3857, 4249, 5731 are placed at the correct hash, these must have been inserted there first in any order. 8131 should belong in index 3, but was found at index 4 so it must have been inserted after 3857 was already there, and linear probing algorithm placed it at index 4. 6201 should belong to index 9, but was found at index 0. It must have been inserted after 4249 was already there. Linear probing algorithm wrapped around the indices back to 0 to place it at index 0.	1m – work out the hash numbers 1m – explain discrepancy for 8131 1m – explain discrepancy for 6201

3(a)	3 C A 7 9 D 4 2 0011 1100 1010 0111 1001 1101 0100 0010	1m – method 1m – ans
(b)	X-axis: 0011 1100 1010 Y-axis: 0111 1001 1101	1m 1m
(c)	X-axis value = $2 + 8 + 64 + 128 + 256 + 512 = 970_{10}$ Y-axis value = $1 + 4 + 8 + 16 + 128 + 256 + 512 + 1024 = 1949_{10}$	1m 1m
(d)	From 0100 0010, the buttons pressed are the B and the Home button.	1m – B 1m – Home

4(a)	 <pre> graph LR Customer[Customer] ===> Booking[Booking] Booking[Booking] ===> Villa[Villa] </pre>	1m – 3 correct attributes 1m - two 1-to-many relations pointing in the correct directions.
4(b)	CUSTOMER (<u>ID</u>, name, email) VILLA (<u>ID</u>, name, cost) BOOKING (<u>ID</u>, <u>customerID</u>, <u>villaID</u>, startDate, numberOfDays)	0m – 3 table descriptions 1m – all correct PK 1m – all correct FK 1m – all fields single word.

5(a)	APPLE, PEACH, ORANGE, BANANA, RAISIN, TOMATO, PLUM	2 marks - deduct 1m per error
5(b)	The nodes will be arranged in alphabetical order from A to Z.	1 mark
5(c)	1. Starts at the root node of the BST and set it as the current node. 2. Compare the value of the item with the value of the current node. (i) Case 1: Match Found if item matches the value of the current node. Return True. (ii) Case 2: Search Left if item is smaller than the value of the current node. Update the current node to its left child. (iii) Case 3: Search Right if item is larger than the value of the current node. Update the current node to its right child. Repeat step 2 until the current node is pointing to null (or is empty). Return False.	1m – start with root node 1m – return True if matches 1m – search left is smaller 1m – search right is larger 1m – repeat until null and return False.

6(a)	Any 2 of the following: (i) Data loss due to misplace or theft of storage device (ii) Data loss due to damage of storage device (iii) prone to malware infection (iv) Unauthorised access to the data (v) Unable to collaborate or share data in realtime.	1 mark each
6(b)	Any 2 of the following: (i) network printer or other peripherals (like scanners or plotters) can be shared (ii) internet access can be shared with a single connection to the Internet Service Provider (ISP) (iii) multi-user software applications can be installed on a central Application Server, allowing all licensed employees to run the program from their individual workstations without installing a copy on every machine. (iv) a local mail servers or internal instant messaging applications can be installed and use within the LAN. (v) Centralised backup of data of storage device (vi) Centralised implementation of network security like authentication, firewall and antivirus software.	1 mark each
6(c)	Any 2 of the following: (i) Cost Efficiency - eliminate the large upfront costs of purchasing physical hardware (servers, storage, network gear)	1 mark each

	- only pay for the resources you actually use, this avoids the waste of unused capacity. - Economies of Scale: small companies can access massive resources and facilities of the Cloud providers - No maintenance of hardware, the cloud provider handles all the costs associated with power, cooling, physical security, and hardware replacement/maintenance. (ii) Scalability: Resources (compute power, storage, bandwidth) can be scaled up or down instantly and automatically to meet fluctuating demand, seasonal spikes, or business growth. (iii) Faster Time-to-Market: Companies can have the Cloud facilities/resources immediately without the waiting time to procure the hardware and setup. (iv) High Availability: Cloud infrastructure typically provide redundancy and high uptime. (v) Provide network and data security. (vi) Accessibility: Data, applications, and services can be accessed from virtually any location, on any device, with an internet connection.	
6(d)	Any 1 of the following services and describe a relevant benefit: Cloud services providers offer a vast array of services far beyond basic file storage: 1. Infrastructure as a Service (IaaS) - eg Networking: Virtual networks, load balancers, firewalls, and DNS services 2. Platform as a Service (PaaS) - eg Databases and Web or App Hosting 3. Software as a Service (SaaS) - eg Productivity Suites: Web-based applications for email, word processing, spreadsheets, and presentations (e.g. Microsoft Office 365, Google Workspace).	1m for stating an online service 1m for describing a relevant benefit.